

MAP-MONO-EGO: MAP-GROUNDED GLOBAL HUMAN POSE ESTIMATION FROM MONOCULAR EGOCENTRIC VIDEO

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ABSTRACT

Monocular egocentric human pose estimation is essential for ubiquitous activity monitoring. However, understanding the user’s absolute location within the environment remains a challenge. Existing methods primarily focus on relative motion from an initial position, and tend not to account for the wearer’s absolute location within an environment. Furthermore, inherent scale ambiguity in monocular vision leads to severe translational drift, limiting long-term tracking without specialized multi-sensor hardware. To address this, we propose Map-Mono-Ego, a novel framework achieving globally consistent pose estimation solely from a monocular camera by leveraging a pre-scanned 3D point cloud. We also introduce AIST-Living dataset, a new dataset pairing egocentric video with ground-truth motion in a scanned environment. Experiments demonstrate that our approach significantly outperforms the state-of-the-art baseline, proving its utility for practical monitoring tasks without specialized hardware.

Index Terms— 3D human pose estimation, egocentric vision, 3D computer vision

1. INTRODUCTION

Estimating human pose using only a lightweight monocular wearable camera, which is common and minimal sensing setting, opens up scalable possibilities for AR/VR and ubiquitous activity monitoring. To realize context aware applications, it is essential to understand not only the user’s body posture, but also their spatial relationship with the surrounding environment. In this paper, we propose Map-Mono-Ego, the framework that achieves spatially consistent human pose estimation from a monocular egocentric camera by leveraging a 3D map pre-scanned by a terrestrial laser scanner as a geometric prior. Fig. 1 shows the overview of Map-Mono-Ego.

Despite its potential, current egocentric pose estimation methods focus on recovering relative body motion within a local coordinate system, typically initialized at the user’s starting position [1, 2]. Consequently, these approaches often fail to account for the wearer’s absolute localization within a global map and lack geometric consistency with the environmental structure. Furthermore, accurate trajectory estimation

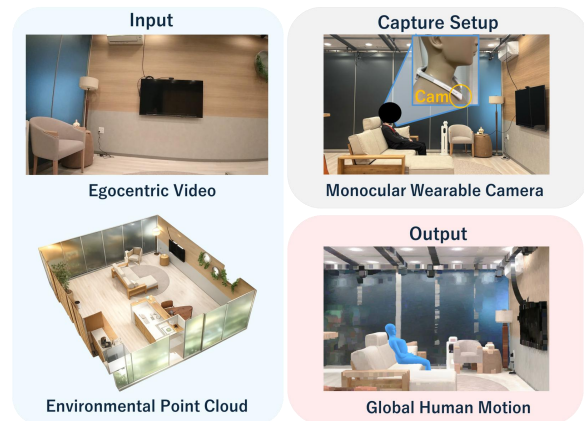


Fig. 1: Overview of our proposed Map-Mono-Ego.

of commodity monocular wearable camera is inherently difficult due to scale ambiguity and motion blur, which leads to severe accumulation of translational errors over time [1, 3]. While specialized multi-sensor hardware can mitigate these issues, reliance on them limits the applicability.

To this end, we focus on monitoring scenarios in controlled environments, such as factories, offices, and residential spaces, where pre-scanning is feasible. In this paper, we propose a framework that leverages high-density 3D scanned point clouds as a geometric prior for estimating human pose. By referencing 3D geometry, our framework recovers a drift-mitigated, metric scale camera trajectory from monocular video, enabling precise global pose estimation. Our contributions are summarized as follows: 1) We propose a framework that estimates globally consistent human pose solely from a monocular egocentric camera by leveraging environmental geometry as a prior. 2) We introduce a robust trajectory tracking algorithm that fuses Hierarchical Localization [4] (HLoc) and Simultaneous Localization and Mapping [3] (SLAM), designed to incorporate environmental priors. 3) We constructed a new benchmark dataset comprising egocentric video, an environmental point cloud, and ground-truth motion data. Experiments on this dataset demonstrate the effectiveness of our method over the state-of-the-art baseline. We release the dataset and annotations at project page.

2. RELATED WORKS

2.1. Human Motion Estimation from Egocentric Video

Capturing human motion with wearable sensors has gained interest in various fields of application. Unlike traditional motion capture systems that consist of multiple external cameras, wearable sensor-based approaches don't require costly equipment and are free from spatial restrictions.

Among these, motion estimation solely from egocentric video with a front-facing camera device has gained attention due to its unique capability to capture environmental interactions and the ubiquity of consumer devices. Since the user's body is often invisible in front-facing views, early approaches relied on interaction cues [5] or control policies [6, 7] to infer pose. However, a significant paradigm shift occurred with the introduction of EgoEgo [1], which demonstrated that head movement could be a condition for full-body pose. Following this, the field has largely shifted toward diffusion-based architectures conditioned on device trajectory [1, 8, 9, 2]. This trend was accelerated by the availability of the standardized devices like Aria Glass [10], and large-scale datasets containing paired egocentric video and motion ground-truth [11].

Despite these advancements, two critical limitations remain. First, the performance of these trajectory-conditioned approaches is heavily dependent on the accuracy of the input camera pose [1]. While high-end devices like Aria Glass provide robust localization, deriving accurate trajectories from standard monocular cameras remains challenging due to scale ambiguity and drift. Second, prior research has focused on evaluating relative motion starting from an initial position. Consequently, evaluation and discussion regarding global human placement within the world coordinate system have not been conducted in previous studies.

In this work, we leverage 3D environmental point clouds to mitigate monocular drift and achieve spatially consistent human pose estimation.

2.2. Camera Pose Estimation

Estimating the 6-DoF camera trajectory is fundamental for egocentric human pose estimation. While SLAM is traditionally adopted for this purpose, relying solely on a monocular camera still primarily suffers from scale ambiguity and accumulated drift caused by rapid motion and blur [3].

To address scale ambiguity, recent approaches often leverage multi-modal sensors like Aria Glass [10]. While effective, relying on specialized hardware limits the scalability to consumer-grade devices. Alternatively, EgoEgo [1] attempts to resolve scale from monocular video by learning the relationship between distance scale and optical flow. However, these methods often struggle to maintain geometric consistency over long sequences. Although standard SLAM systems incorporate loop closure to mitigate drift, they are often

insufficient in scenarios where the user does not revisit previous locations frequently or in texture-less environments [3].

To address these limitations, we integrate HLoc [4] into the pipeline. Unlike pure SLAM or learning-based odometry, HLoc, which is a structure-based localization method [12], leverages a pre-built 3D map to establish robust 2D-3D correspondences. By using these absolute poses as global anchors, our method ensures drift-mitigated, metric scale tracking using only a monocular camera, providing a reliable foundation for human motion estimation.

3. METHOD

Our goal is to recover the global human motion sequence X from T frames of an egocentric video $I = \{I_t\}_{t=1}^T$, and a pre-scanned 3D point cloud P^{scan} . As illustrated in Fig. 2, Map-Mono-Ego operates in three stages: ① **Localization via Synthetic Database**: Estimating camera poses initially by matching the video frames against a synthetically rendered database from the point cloud. ② **Trajectory Refinement by Inlier-based Filtering**: Filtering for reliable camera poses based on geometric consistency and interpolating the trajectory using SLAM to ensure smoothness and global consistency. ③ **Diffusion-based Human Pose Estimation**: Predicting the human motion sequence using a diffusion model conditioned on the refined trajectory. By feeding the robust trajectory from ① and ② into the motion diffusion pipeline of ③, our framework achieves spatially consistent monocular egocentric human pose estimation.

3.1. Localization via Synthetic Database

To estimate the camera poses within the scanned environment, we initially employ HLoc [4]. Since HLoc requires an image-to-geometry reference, we construct N frames of synthetic database $\mathcal{D} = \{I_n^{db}, \mathbf{P}_n^{db}, \mathcal{C}_n\}_{n=1}^N$ from P^{scan} . Specifically, we generate synthetic views and ground-truth poses from P^{scan} following the protocol in HPS [13]. Each entry contains the synthetic image I_n^{db} , its pose $\mathbf{P}_n^{db} \in \text{SE}(3)$, and 3D correspondences $\mathcal{C}_n$. We then treat the input video frames $\mathcal{I}$ as queries. By performing visual localization against $\mathcal{D}$, we obtain the initial camera pose $\hat{\mathbf{P}}_t^{loc} \in \text{SE}(3)$ for each frame t .

3.2. Trajectory Refinement by Inlier-based Filtering

The raw poses $\hat{\mathbf{P}}_t^{loc}$ may contain outliers due to motion blur or textureless regions. To ensure reliability, we filter these poses based on the PnP inlier count and ratio derived from aggregated matches against the top-40 database candidates retrieved by a global descriptor in HLoc. As visualized in Fig. 3, these metrics correlate strongly with localization accuracy. We retain only the poses where both the inlier count and ratio exceed specific thresholds, defining a subset of reliable indices $\mathcal{K} \subset \{1, \dots, T\}$.

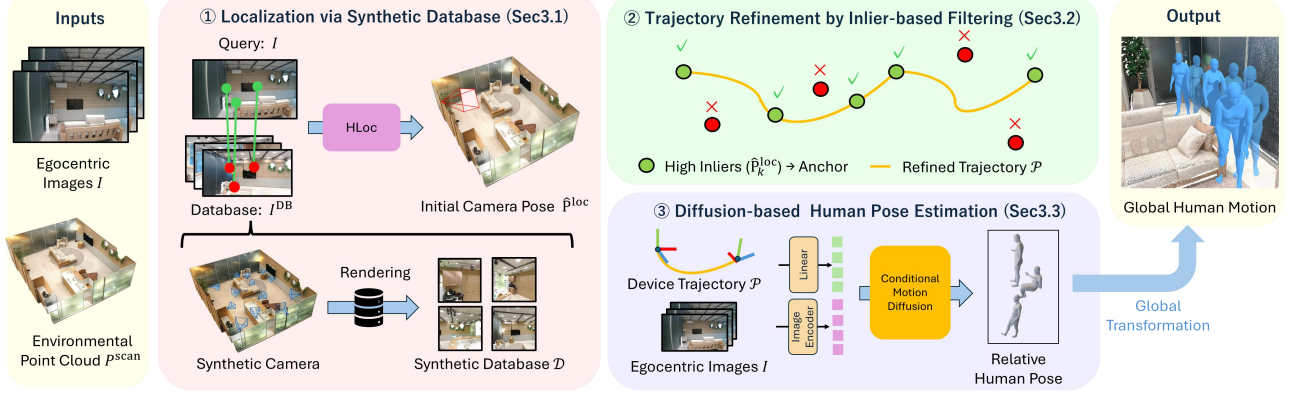


Fig. 2: Overview of the Map-Mono-Ego framework. ① We estimate initial camera pose $\hat{\mathbf{P}}^{\text{loc}}$ (Sec. 3.1). ② We filter $\hat{\mathbf{P}}^{\text{loc}}$ for reliable camera poses and recover a smooth, drift-free trajectory $\mathcal{P}$ via SLAM-based interpolation (Sec. 3.2). ③ We predict global human motion using the refined trajectory. $\mathcal{P}$ (Sec. 3.3).



Fig. 3: The left columns visualize matches between egocentric images and top-1 retrieved images. The right columns compare real egocentric views with synthetic views, which render point cloud P^{scan} from estimated poses. High inlier metrics (top) demonstrate precise alignment, while low metrics (bottom) reveal noticeable misalignment.

To reconstruct the full trajectory $\mathcal{P} = \{\mathbf{P}_t\}_{t=1}^T$, we interpolate between the reliable anchor poses $\{\hat{\mathbf{P}}_k^{\text{loc}}\}_{k \in \mathcal{K}}$ by aligning the continuous trajectory estimated in a local coordinate frame by DROID-SLAM [14]. Consider an interval between two consecutive reliable frames $n, m \in \mathcal{K}$ ($n < m$) the corresponding SLAM pose sequence $\{\mathbf{P}_t^{\text{slam}}\}_{t=n}^m$. First, we compute the similarity transformation $S \in \text{Sim}(3)$ that aligns the SLAM defined frame to the world frame at the start frame n :

$$S = \hat{\mathbf{P}}_n^{\text{loc}} (\mathbf{P}_n^{\text{slam}})^{-1}. \quad (1)$$

To correct the accumulated scale drift inherent in monocular SLAM, we define a residual transformation $E \in \text{Sim}(3)$ that aligns the propagated pose with the reliable anchor at the end frame m :

$$E = \hat{\mathbf{P}}_m^{\text{loc}} (S \mathbf{P}_m^{\text{slam}})^{-1}. \quad (2)$$

We distribute this residual across the interval using a time-dependent factor $\alpha_t = \frac{t-n}{m-n}$. Finally, the refined global camera pose $\mathbf{P}_t$ is computed via Lie algebra interpolation [15]:

$$\mathbf{P}_t = \exp(\alpha_t \log(E)) S \mathbf{P}_t^{\text{slam}}. \quad (3)$$

This formulation ensures the trajectory strictly satisfies boundary constraints at n and m while preserving the local geometric structure captured by SLAM.

3.3. Diffusion-based Human Pose Estimation

Finally, we estimate the SMPL-X [16] motion sequence $X = \{\{\mathbf{R}_t^{\text{root}}, \mathbf{t}_t^{\text{root}}, \theta_t\}_{t=1}^T, \beta\}$ using a Transformer-based diffusion model following UniEgoMotion [2]. Here, $\mathbf{R}_t^{\text{root}} \in \mathbb{R}^3$ and $\mathbf{t}_t^{\text{root}} \in \mathbb{R}^3$ denote the root joint's global rotation and translation, $\theta_t \in \mathbb{R}^{21 \times 3}$ denotes the local joint angles excluding hands and face, and $\beta \in \mathbb{R}^{10}$ represents the time invariant body shape. Unlike UniEgoMotion [2], which targets head-mounted devices, we use a neck-mounted camera (as shown in Fig. 1) [17] in our experiment. Therefore, we retrained the model to accept the neck joint trajectory as the condition.

Before the inference, we transform the refined camera trajectory $\mathcal{P}$ into a canonicalized trajectory $\mathcal{P}^{\text{cano}}$ where the first frame is centered at the origin and aligned with the forward axis. The architecture iteratively recovers X from Gaussian noise, conditioned on the canonicalized camera trajectory $\mathcal{P}^{\text{cano}}$ and DINOv2 [18] image features. Finally, we transform the predicted relative motion back to the world frame by applying the inverse canonicalization to $\mathbf{R}^{\text{root}}$ and $\mathbf{t}^{\text{root}}$.

4. EXPERIMENTS

4.1. Dataset

To train the motion diffusion model, we use EE4D-motion dataset [2]. Following UniEgoMotion [2], we trained on 8-second videos at 10fps. On the other hand, for benchmarking, a dataset pairing environmental point clouds, egocentric video, and ground-truth motion data was required. Therefore, we constructed AIST-Living dataset. We collected the data in

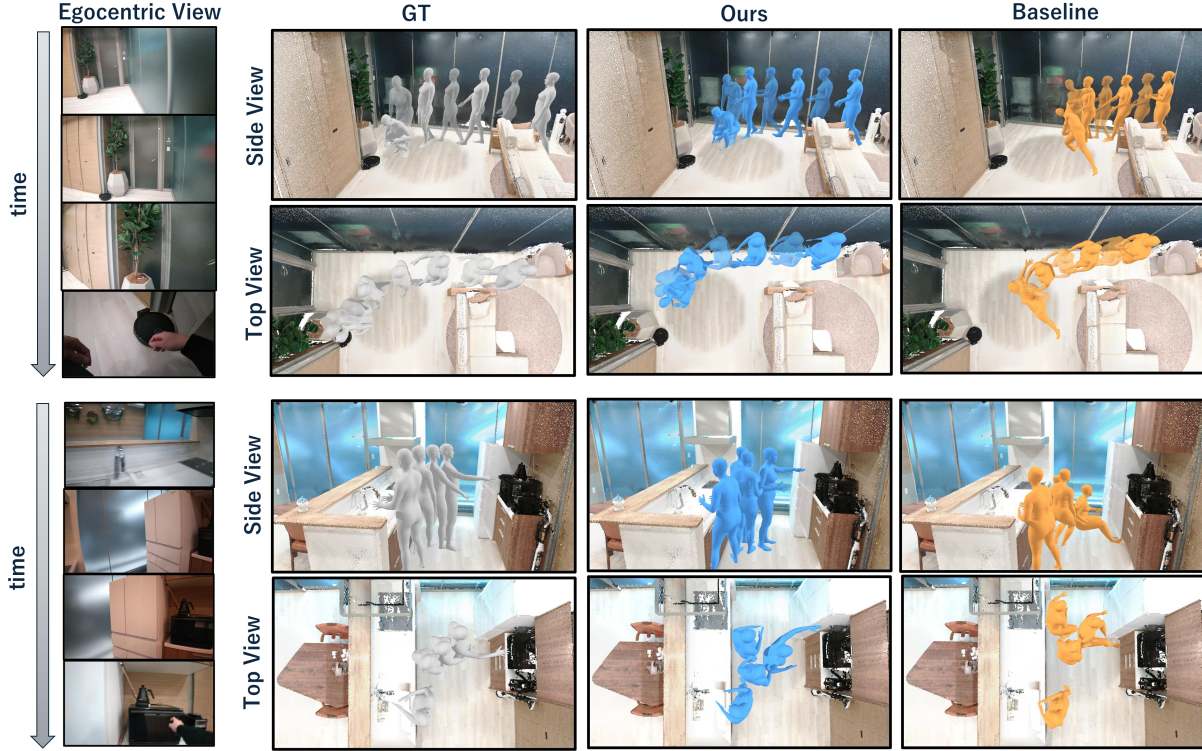


Fig. 4: Qualitative comparison of global human pose estimation. The top one shows a sequence where the subject walks to a robotic vacuum cleaner and crouches down. The bottom one shows a sequence where the subject walks to a microwave in the kitchen and reaches for it. Our method estimates more precise and natural motion than the baseline in both sequences.

a laboratory environment designed to simulate a typical living room, assuming a real-world scenario of daily activity monitoring. AIST-Living dataset comprises 152 sequences of 8 seconds each at 10 FPS, and we used it for evaluation. For more details, please refer to the supplementary material.

4.2. Baseline

To validate the benefit of environmental priors within the scalable monocular setting (avoiding reliance on high-end sensors like Aria glass [10, 8, 9]), we compare against a method relying solely on monocular vision. We adopt the trajectory estimation algorithm from EgoEgo [1], the state-of-the-art method specifically designed for the monocular egocentric human pose estimation, and refer to it as our baseline.

EgoEgo employs a hybrid approach that integrates learning-based algorithms with monocular SLAM to achieve metric scale camera trajectory estimation. Specifically, it combines DROID-SLAM [14] for trajectory tracking with GravityNet for gravity alignment and HeadNet, which processes optical flow features extracted by RAFT [19] and ResNet-18 [20], to estimate the metric scale translation and rotation of a monocular camera. In our implementation, we utilize the official pretrained models for both GravityNet and HeadNet. To isolate the impact of the trajectory estimation method, we feed

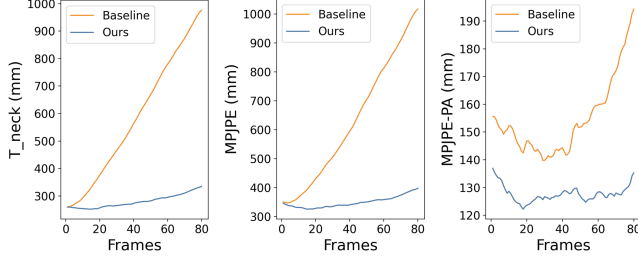
the metric-scale trajectory obtained by this EgoEgo-based framework into the same motion diffusion model used in Map-Mono-Ego. Furthermore, since the baseline operates in a local coordinate system relative to the start frame, we align its initial global position and orientation using the results of our method to ensure a fair comparison.

4.3. Evaluation Metrics

To assess neck-mounted camera tracking accuracy, we report the Neck Orientation Error (O_{neck}) and the Translation Error (T_{neck}), measured in mm. The orientation error is computed as the Frobenius norm of the rotation matrix. For pose accuracy, we report standard protocols, all measured in mm. **MPJPE** computes the mean per-joint positional error over 22 body joints. **MPJPE-Rigid** further aligns the predicted motion to the ground truth using a single rigid transformation *per sequence* to remove a global offset, and measures *sequence-level* motion consistency. **MPJPE-PA** applies Procrustes analysis *per frame* to align the predicted and ground-truth motions, measuring the accuracy of *frame-level* local pose predictions. To evaluate physical realism, we report Foot Sliding (**FS**) [21] and Foot Contact (**FC**), both measured in mm. Specifically, FS quantifies the sliding distance when the foot is close to the ground, while FC computes the average

Table 1: Quantitative comparison. Bold numbers denote the better performance for each metric.

	$O_{\text{neck}} \downarrow$	$T_{\text{neck}} \downarrow$	MPJPE $\downarrow$	MPJPE-Rigid $\downarrow$	MPJPE-PA $\downarrow$	FS $\downarrow$	FC $\downarrow$	SS $\uparrow$
Baseline	0.66	577.3	623.0	351.0	152.9	16.0	7.8	0.76
Ours	0.66	279.7	347.5	201.3	128.6	11.4	2.2	0.82

**Fig. 5:** We plot the average T_{neck} , MPJPE, and MPJPE-PA errors at each frame index across all sequences. Our method achieves drift-mitigated human motion estimation.

foot-ground separation to capture floating and penetration artifacts. Finally, we report Semantic Similarity (SS), which evaluates perceptual motion quality in a semantic latent space via the TMR [22] motion encoder.

4.4. Comparison with the Baseline

As summarized in Table 1, Fig. 4, and Fig. 5, we evaluate our pipeline against the baseline on AIST-Living dataset.

As shown in Table 1, our method achieves significantly more accurate results in T_{neck} and competitive performance in O_{neck} . This confirms that our map-grounded approach estimates a more precise camera trajectory than the baseline. Crucially, our method consistently outperforms the baseline across all remaining pose metrics, including MPJPE-Rigid, MPJPE-PA, and Semantic Similarity (SS). The poor performance of the baseline on these metrics suggests a fundamental issue in the motion estimation process. We attribute this to the failure of the baseline in estimating the correct metric scale and gravity alignment. Since the diffusion model is conditioned on the neck trajectory, an input trajectory with incorrect translational scale or gravity alignment acts as a physically inconsistent condition. It makes the motion diffusion model output an unnatural body posture, thereby degrading not only global positioning but also the local fidelity and semantic quality of the estimated motion.

Qualitative comparisons in Fig. 4 further validate these findings. Our method successfully achieves precise global human motion, accurately reconstructing interactions with the environment, such as sitting near a robot vacuum or reaching for a microwave. In contrast, the baseline predicts unnatural motions characterized by significant translational drift and erroneous neck heights. It shows that monocular methods struggle to estimate plausible human-scene interaction without environmental priors.

Furthermore, frame-wise analysis in Fig. 5 clearly demon-

Table 2: Ablation study for the effects of our trajectory refinement ②. Values denoted by '—' indicate that the error exceeded 10^4 mm, representing a failure in the estimation.

	$O_{\text{neck}} \downarrow$	$T_{\text{neck}} \downarrow$	MPJPE $\downarrow$
Ours	0.66	279.7	347.5
Ours w/o ②	1.41	—	—

strates that the baseline suffers from a severe accumulation of errors in T_{neck} and MPJPE over time, exhibiting the characteristic effect of monocular drift. Notably, the baseline’s local pose fidelity (MPJPE-PA) also degrades in later frames, whereas ours remains stable. This stability highlights how robust trajectory tracking prevents the diffusion model from generating implausible poses over long sequences.

4.5. Ablation Study

We studied the effects of our trajectory refinement using several metrics in Table 2. Without this refinement, raw HLoc localization suffers from numerous outliers, leading to inaccurate and discontinuous camera trajectories. It leads to a substantial increase in MPJPE, demonstrating the necessity of our refinement process for stable global pose estimation.

5. CONCLUSION

In this study, we propose Map-Mono-Ego, the framework that effectively utilizes environmental point clouds and monocular egocentric video to estimate the global human pose. Specifically, we leverage environmental point clouds as geometric priors through HLoc-based localization and inlier-based trajectory refinement. By integrating this robust tracking into a diffusion framework, we realize spatially consistent motion estimation. Experiments demonstrate that our method significantly outperforms the state-of-the-art baseline method, and shows its utility for practical monitoring applications in daily environments. For future work, we aim to explicitly incorporate the 3D scene geometry into the motion inference process to predict more physically plausible motions, including plausible human-object and human-scene interactions.

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